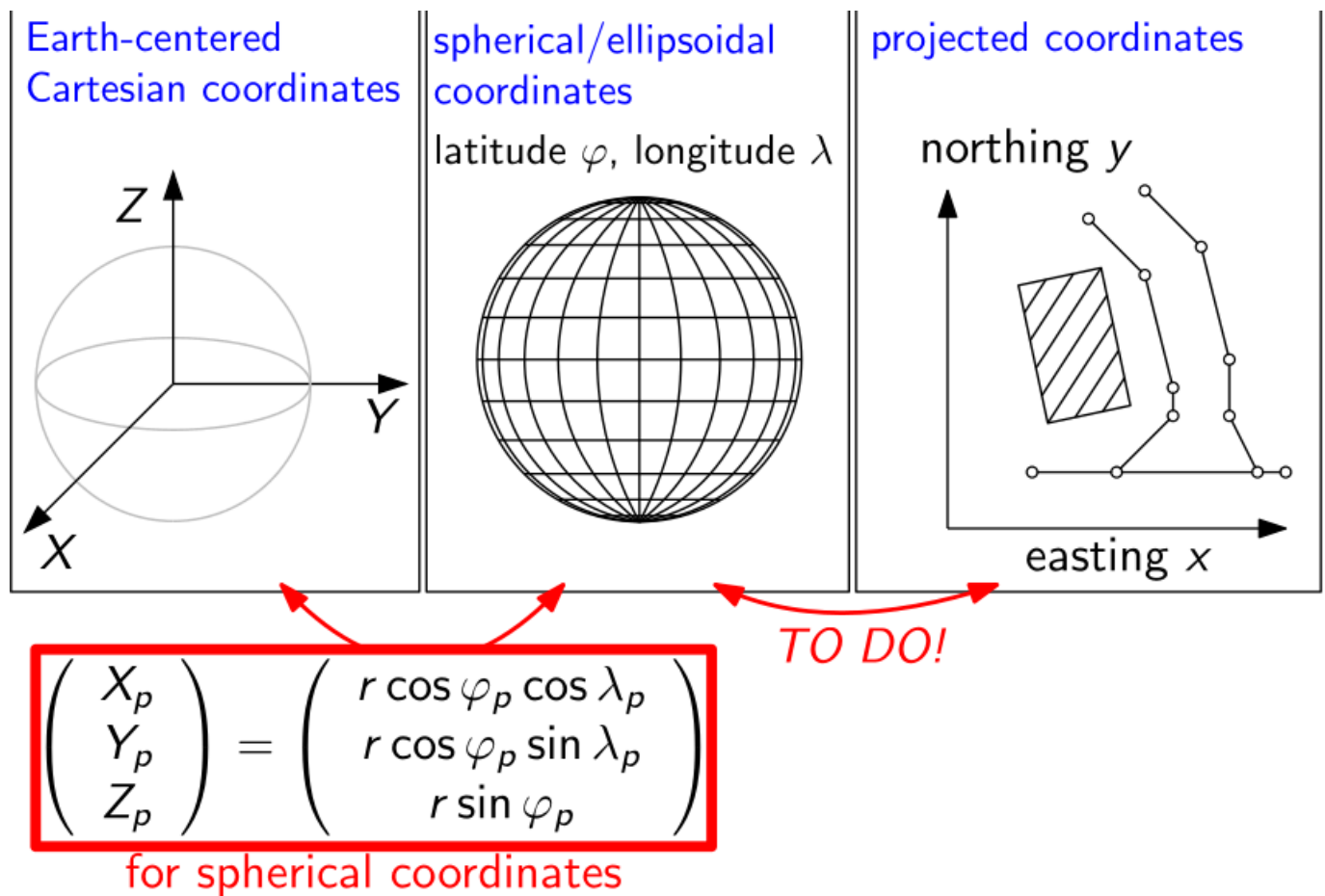
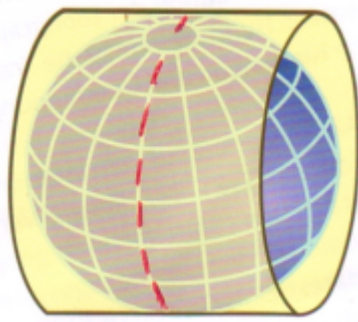


L7: Map Projections



Cylindrical Projection

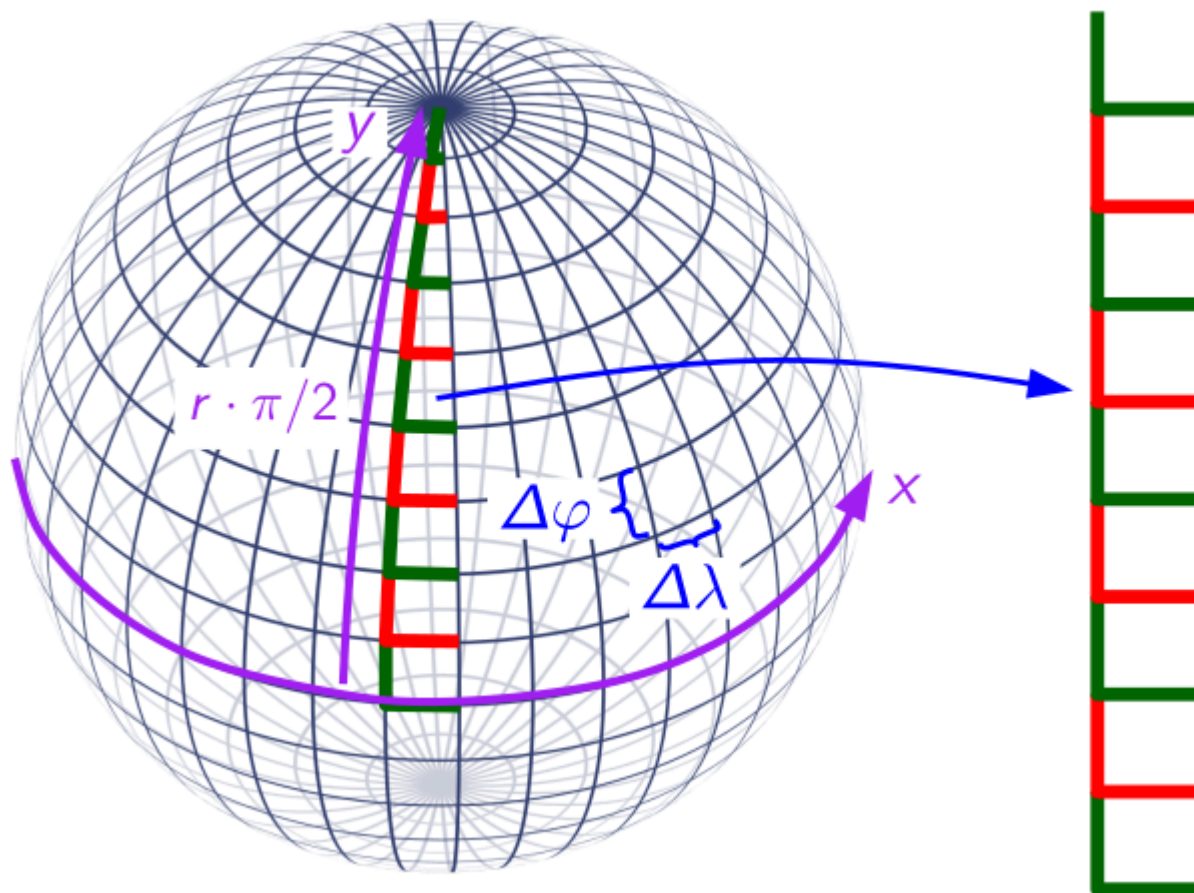
- e.g. Plate Carrée
- For **all** cylindrical projections in normal position:
 - Lines of equal latitude are mapped to horizontal lines
 - Meridians are mapped to vertical lines
 - Equator scaled with constant factor (often 1)
- Distances between lines of equal latitude can be non-uniform
- For *transverse* cylindrical projections, consider one meridian as the Earth's equator
 - Mercator projection



o

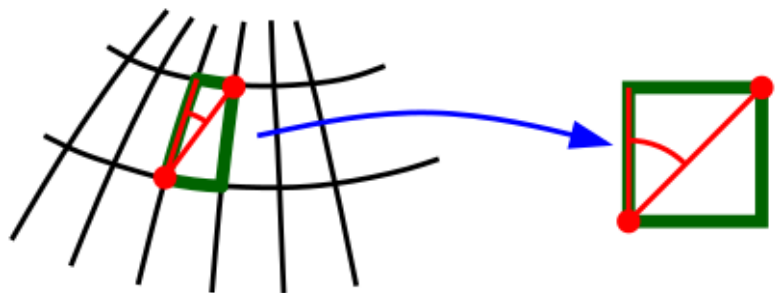
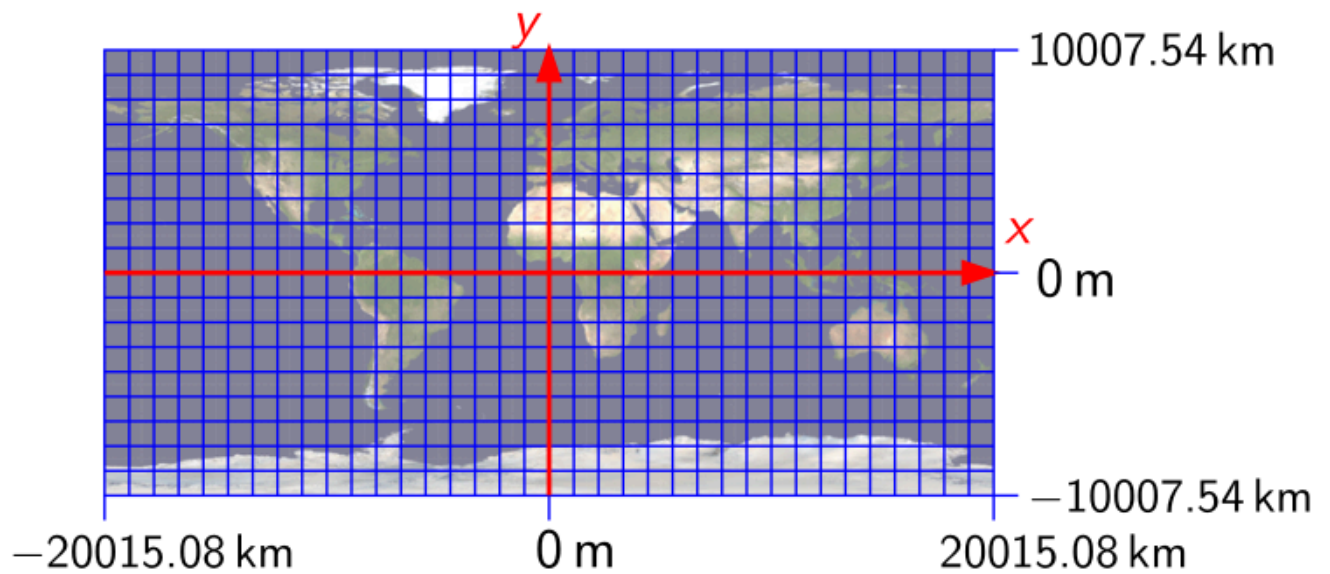
Plate Carrée Projection

- Map areas between lines of constant latitude/longitude to **squares of constant size** (assuming $\Delta\varphi = \Delta\lambda$)
- Preserve lengths of equator and meridians



-
- **Mapping Function:**
 - o Easting $x = \lambda_{deg} \cdot \frac{\pi}{180^\circ} \cdot r$
 - o Northing $y = \varphi_{deg} \cdot \frac{\pi}{180^\circ} \cdot r$
 - o with r = Earth radius

- $\begin{pmatrix} x_p \\ y_p \end{pmatrix} = \begin{pmatrix} r\lambda_p \\ r\lambda_p \end{pmatrix}$



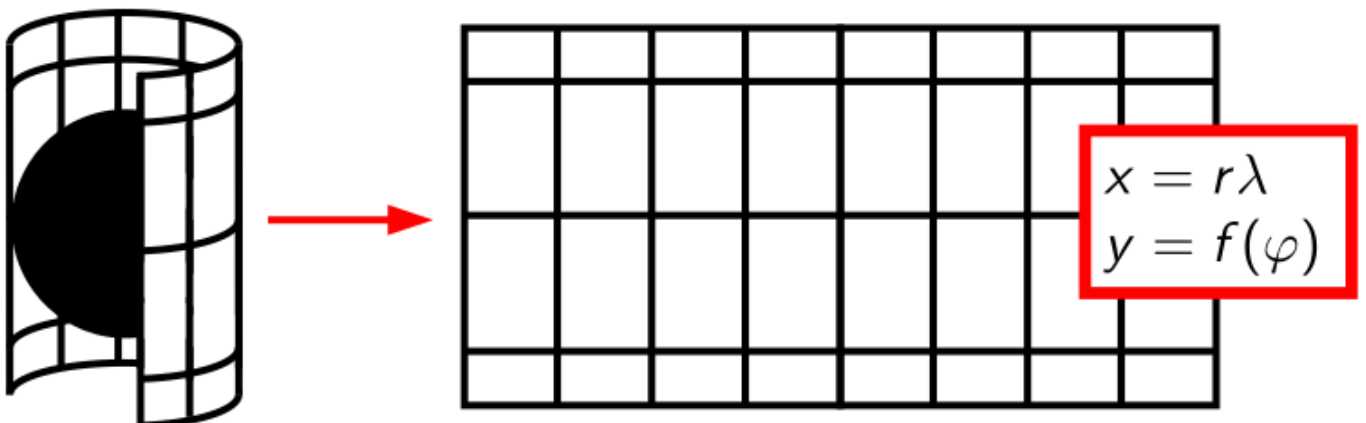
Disadvantages

- **Shapes** get "Squeezed", i.e. aspect ratios change
- **Angles** change
- **Areas** Change

Advantages

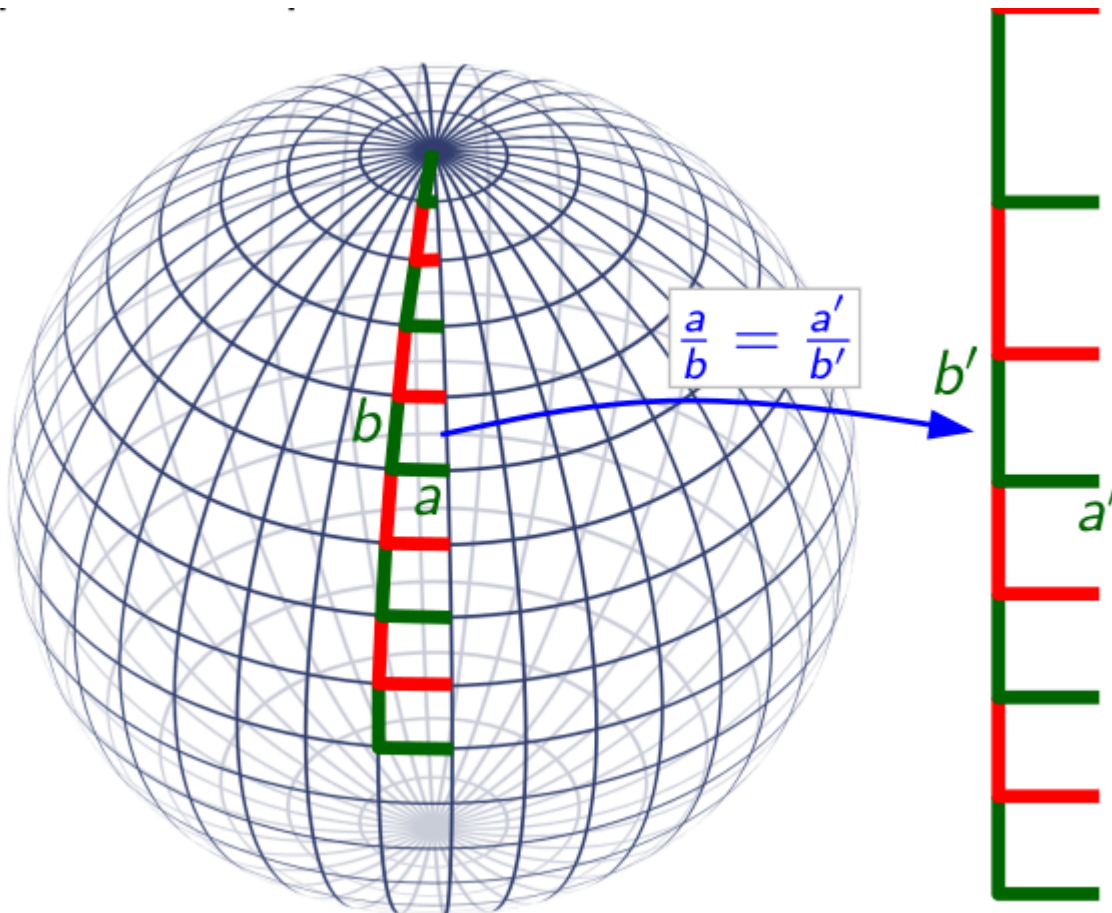
- **Lengths of equator and meridians** are preserved

For all cylindrical projections in normal position:



Mercator Projection

- Map areas between lines of constant latitude/longitude to rectangles that preserve aspect ratios
- Here shown for $\Delta\lambda = \Delta\varphi = 10^\circ$
- For exact construction: Choose $\Delta\lambda, \Delta\varphi$ infinitely small



- Mapping Function
 - $x_p = r\lambda_p$
 - $y_p = r \ln \tan\left(\frac{\pi}{4} + \frac{\varphi_p}{2}\right)$
 - r = Earth's radius

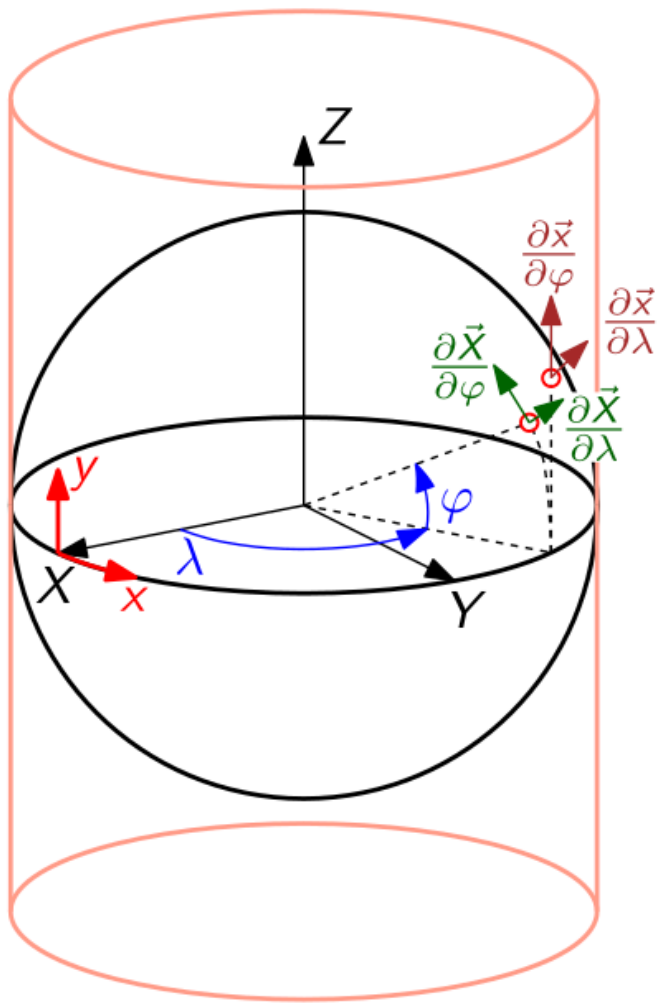
Disadvantages

- Areas are distorted (esp. close to poles)
- Lengths of meridians are not preserved

Advantages

- Aspect ratios are preserved
- (->) Angles are preserved
- Mercator projection is **conformal**

Mercator Mapping function derivation



Requirement:
same aspect ratios

$$\frac{\left| \frac{\partial \vec{x}}{\partial \varphi} \right|}{\left| \frac{\partial \vec{x}}{\partial \lambda} \right|} = \frac{\left| \frac{\partial \vec{X}}{\partial \varphi} \right|}{\left| \frac{\partial \vec{X}}{\partial \lambda} \right|}$$

$$\Leftrightarrow \frac{\left(\frac{\partial \vec{x}}{\partial \varphi} \right)^2}{\left(\frac{\partial \vec{x}}{\partial \lambda} \right)^2} = \frac{\left(\frac{\partial \vec{X}}{\partial \varphi} \right)^2}{\left(\frac{\partial \vec{X}}{\partial \lambda} \right)^2}$$

$$\Leftrightarrow \frac{(f'(\varphi))^2}{r^2} = \frac{r^2}{r^2 \cos^2 \varphi}$$

$$\Leftrightarrow f'(\varphi) = \frac{r}{\cos \varphi}$$

$$\Leftrightarrow f(\varphi) = r \ln \tan\left(\frac{\pi}{4} + \frac{\varphi}{2}\right)$$

$$\left(\frac{\partial \vec{x}}{\partial \varphi}\right)^2 = \begin{pmatrix} 0 \\ f'(\varphi) \end{pmatrix}^2 = (f'(\varphi))^2$$

$$\left(\frac{\partial \vec{x}}{\partial \lambda}\right)^2 = \begin{pmatrix} r \\ 0 \end{pmatrix}^2 = r^2$$

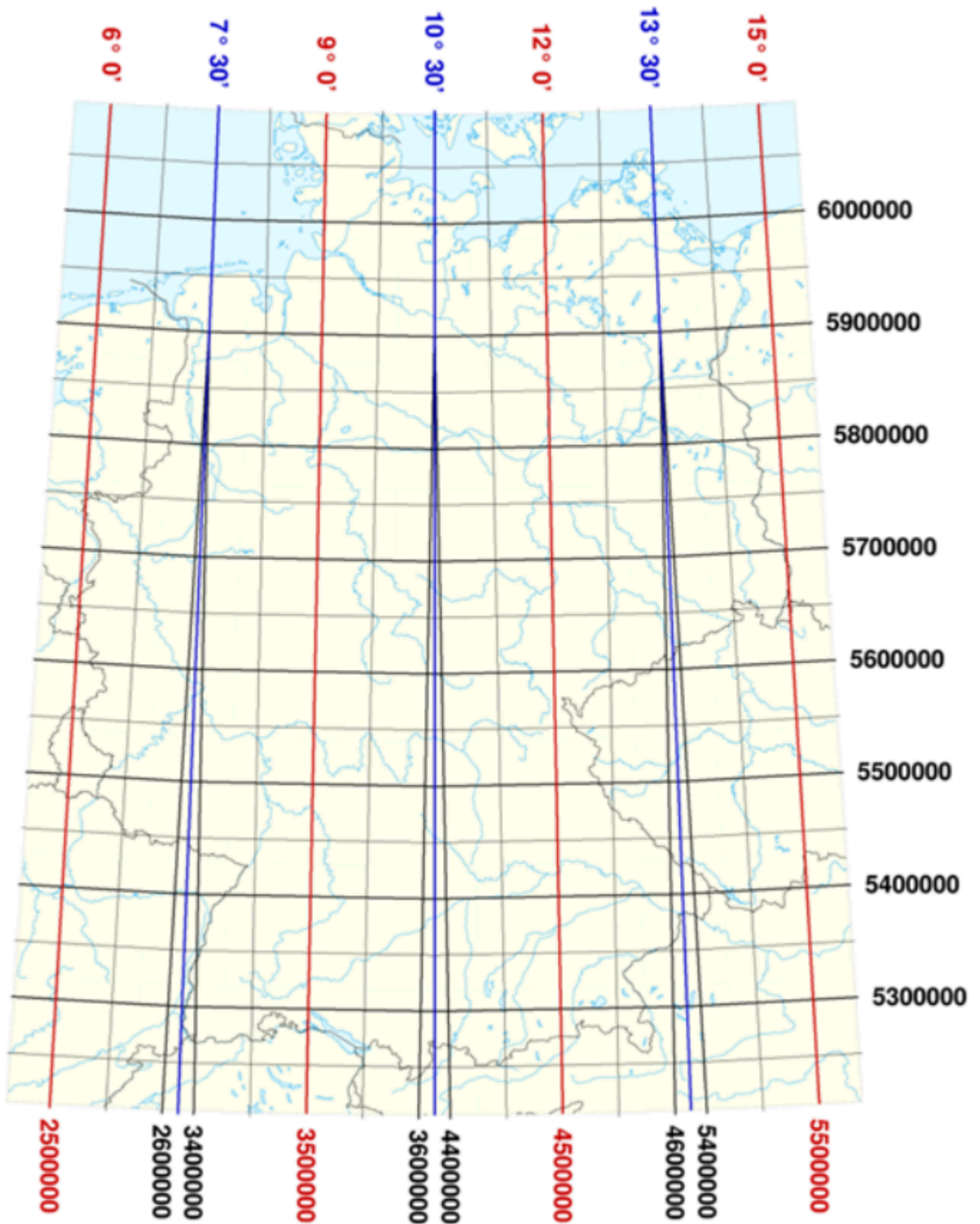
$$\left(\frac{\partial \vec{X}}{\partial \varphi}\right)^2 = \begin{pmatrix} -r \sin \varphi \cos \lambda \\ -r \sin \varphi \sin \lambda \\ r \cos \varphi \end{pmatrix}^2 = r^2$$

$$\left(\frac{\partial \vec{X}}{\partial \lambda}\right)^2 = \begin{pmatrix} -r \cos \varphi \sin \lambda \\ r \cos \varphi \cos \lambda \\ 0 \end{pmatrix}^2 = r^2 \cos^2 \varphi$$

with

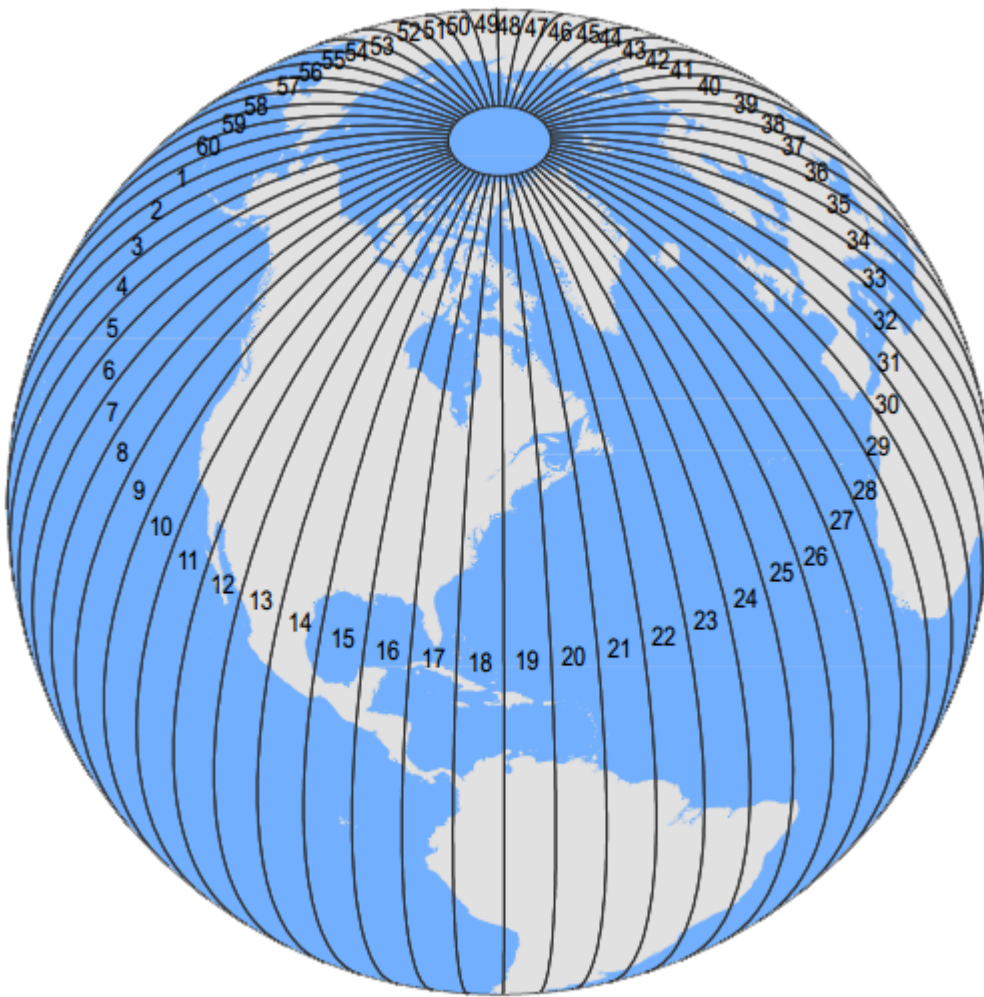
Gauß Krüger projection

- Transversed Cylindrical Projection
- Earth subdivision into 3° width zones
- For different zones, different projections
 - Central meridian = center of zone
- Western German system (replaced by UTM)



Universal Transverse Mercator - UTM

- Similar to Gauß-Krüger
- Width of zone 6°
- Scale=0.9996



- **Mapping function:**

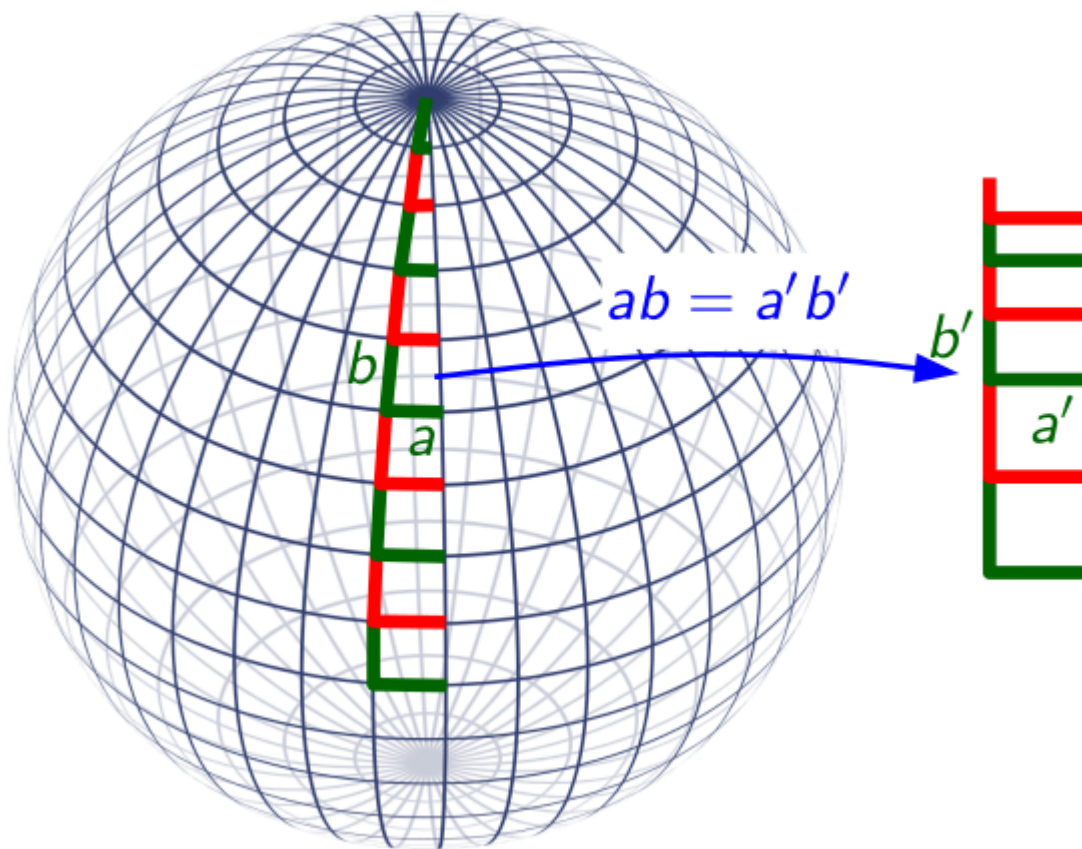
- $x_p = r\lambda_p$
- $y_p = r \ln \tan \left(\frac{\pi}{4} + \frac{\varphi_p}{2} \right)$

Conclusion

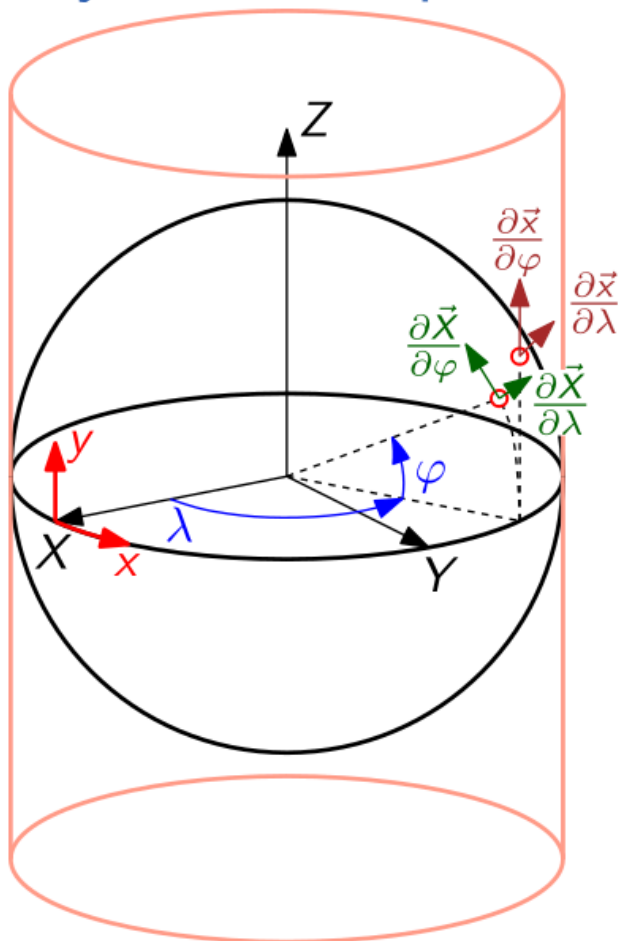
- Mercator projection not appropriate for world maps
- Frequently used for smaller zones
 - Close to equator or
 - Close to a central meridian (transverse version)

Cylindrical Equal-Area Projection

- Map areas between lines of constant latitude/longitude to rectangles of the same sizes
- Here shown for $\Delta\lambda = \Delta\varphi = 10^\circ$
- For exact construction: $\Delta\lambda, \Delta\varphi$ infinitely small



Cylindrical Equal-area Projection – Maths



Requirement:
same sizes

$$\left| \frac{\partial \vec{x}}{\partial \varphi} \right| \cdot \left| \frac{\partial \vec{x}}{\partial \lambda} \right| = \left| \frac{\partial \vec{X}}{\partial \varphi} \right| \cdot \left| \frac{\partial \vec{X}}{\partial \lambda} \right|$$

$$\Leftrightarrow \left(\frac{\partial \vec{x}}{\partial \varphi} \right)^2 \left(\frac{\partial \vec{x}}{\partial \lambda} \right)^2 = \left(\frac{\partial \vec{X}}{\partial \varphi} \right)^2 \left(\frac{\partial \vec{X}}{\partial \lambda} \right)^2$$

$$\Leftrightarrow (f'(\varphi))^2 r^2 = r^4 \cos^2 \varphi$$

$$\Leftrightarrow f'(\varphi) = r \cos \varphi$$

$$\Leftrightarrow \underline{\underline{f(\varphi) = r \sin \varphi}}$$

$f(0) = 0$

• Mapping function:

$$\circ x_p = r \lambda_p$$

- $y_p = r \sin \varphi_p$
 - r = Earth's radius

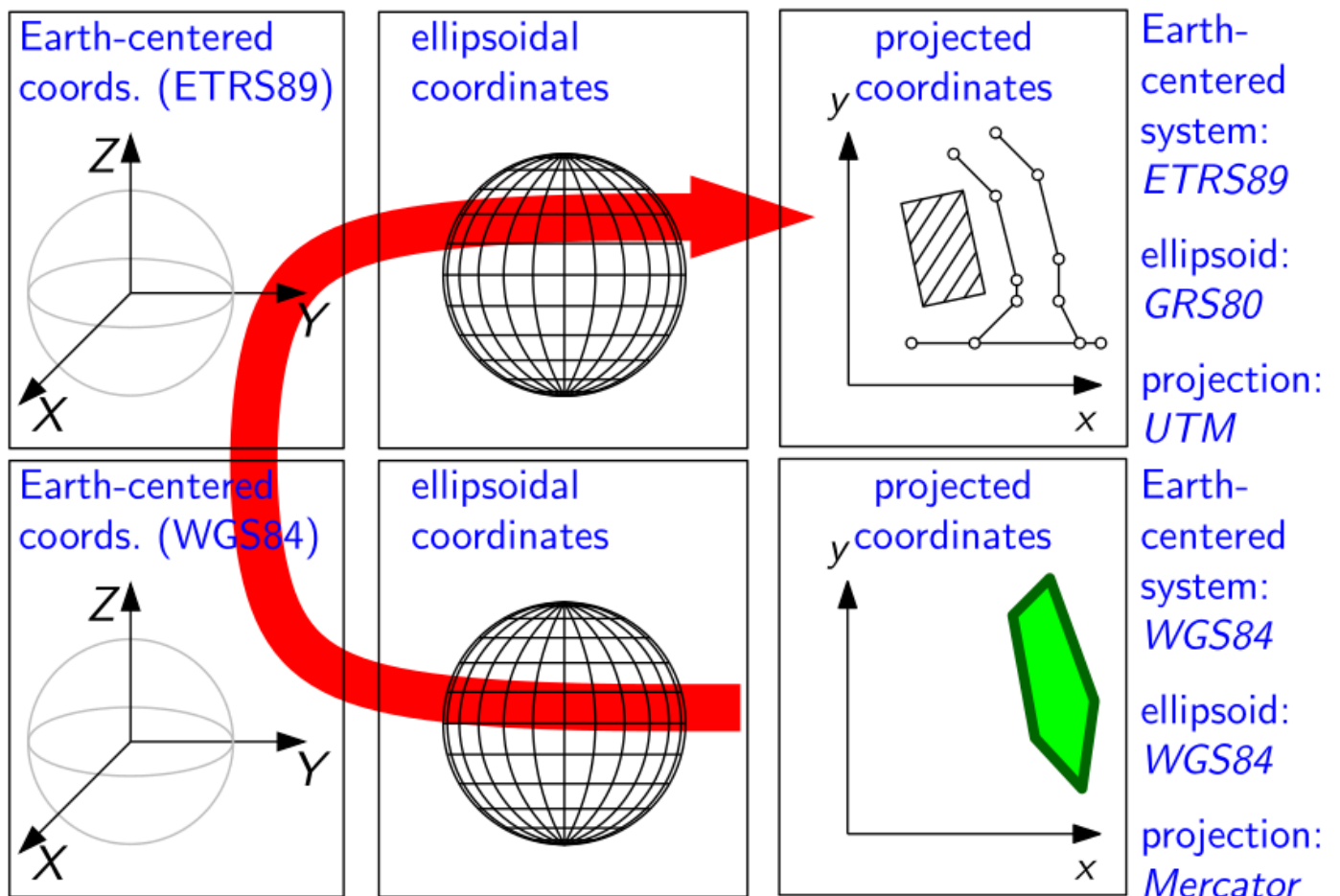
Disadvantages

- Cylindrical equal-area projection leads to large distortions of aspect ratios and angles

More Map Projections

- For whole World: Compromise between area distortion and angle distortion
 - e.g. Robinson projection
- **No map projection preserves both areas and angles!**

Coordinate Transformation



Name	Plate Carrée Projection	Mercator Projection	Cylindrical Equal-Area Projection
Mapping Function	$\text{Easting } x = \lambda_{[deg]} \cdot \frac{\pi}{180^\circ} \cdot r$ $\text{Northing } y = \varphi_{[deg]} \cdot \frac{\pi}{180^\circ} \cdot r$	$x_p = r \lambda_p$ $y_p = r \ln \tan\left(\frac{\pi}{4} + \frac{\varphi_p}{2}\right)$	$x_p = r \lambda_p$ $y_p = r \sin$

Name	Plate Carrée Projection	Mercator Projection	Cylindrical Equal-Area Projection
Disadvantages	aspect ratios change; Angles change; Areas Change	Areas distorted (esp. close to poles); Lengths of meridians are not preserved	Large Distortions of aspect ratio and angles
Advantages	Lengths of equator and meridians are preserved	Aspect ratios preserved; - > Angles preserved, Mercator projection is conformal	Equal Areas
	